

SAN PEDRO SULA MORALES, Honduras

WMO: 787080

Lat: 15.453N

Lon: 87.924W

Elev: 28

StdP: 100.99

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	18.8	15.8	11.2	23.5	16.9	12.1	23.0	7.9	27.2	7.0	27.5	1.2	210	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.1	36.8	25.9	35.6	26.1	34.8	26.2	28.7	32.7	28.2	31.9	27.8	31.4	4.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.1	30.7	27.2	23.1	29.9	27.0	22.8	29.7	93.3	33.1	91.2	31.9	89.3	31.8	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.3	6.4	16.0	40.0	1.1	1.5	15.2	41.1	14.6	42.0	14.0	42.8	13.2	43.9		
			15.3	29.8	1.2	0.8	14.5	30.4	13.8	30.8	13.1	31.2	12.2	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	24.2	25.5	26.5	28.5	29.3	28.9	28.3	28.6	28.8	27.4	25.3	24.7	(6)	
	DBStd	2.48	2.08	2.10	2.41	2.12	1.49	1.19	0.94	0.96	1.02	1.75	2.06	2.00	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6269	441	433	512	556	598	567	567	577	565	539	459	455	(10)	
	CDD18.3	3227	182	200	254	306	339	317	309	318	315	281	209	197	(11)	
	CDH23.3	30867	1108	1537	2298	3346	3894	3487	3222	3396	3402	2545	1398	1234	(12)	
	CDH26.7	13259	359	597	1012	1604	1844	1593	1376	1522	1550	978	440	384	(13)	
Wind	WSAvg	2.3	2.0	2.2	2.7	2.8	2.9	2.5	2.5	2.4	2.4	2.0	1.8	1.9	(14)	
Precipitation	PrecAvg	1192	71	57	38	35	74	167	94	104	163	153	148	107	(15)	
	PrecMax	1630	267	173	114	150	300	355	252	215	549	431	384	246	(16)	
	PrecMin	527	6	1	0	0	7	12	3	16	25	28	16	15	(17)	
	PrecStd	303	52	40	33	38	66	90	49	45	104	92	81	61	(18)	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.1	38.0	39.2	37.9	36.2	35.0	35.8	36.2	35.2	33.9	33.2	(19)	
		MCWB	24.4	24.4	24.0	25.8	26.3	26.3	26.3	26.6	25.9	26.4	25.4	25.2	(20)	
	2%	DB	31.3	33.1	35.1	36.9	36.1	35.1	34.0	34.6	35.1	34.0	32.0	31.5	(21)	
		MCWB	24.7	24.6	24.8	26.0	26.6	26.5	26.3	26.5	26.3	26.6	25.9	25.4	(22)	
	5%	DB	30.1	31.9	33.6	35.0	35.0	34.1	33.1	33.8	34.1	32.9	30.9	30.1	(23)	
		MCWB	24.6	24.7	25.0	26.1	26.4	26.5	26.2	26.6	26.5	26.4	25.7	25.2	(24)	
	10%	DB	28.9	30.2	31.9	33.5	33.9	33.1	32.2	32.9	33.1	31.8	29.2	29.0	(25)	
		MCWB	24.3	24.5	24.7	25.9	26.3	26.4	26.3	26.4	26.4	26.2	25.4	24.8	(26)	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	27.0	28.1	29.2	29.2	29.1	28.7	28.8	29.0	28.5	27.8	27.5	(27)
			MCDB	29.9	31.4	32.6	33.8	34.0	33.0	32.0	32.6	33.0	32.1	30.4	30.5	(28)
2%		WB	25.9	26.1	26.7	28.1	28.2	28.3	28.0	28.1	28.2	27.8	27.0	26.5	(29)	
		MCDB	29.1	30.1	31.1	32.5	32.4	31.8	31.2	31.7	31.9	31.1	30.0	29.4	(30)	
5%		WB	25.2	25.6	26.1	27.3	27.6	27.7	27.5	27.6	27.7	27.2	26.3	25.9	(31)	
		MCDB	28.5	29.3	30.2	31.6	31.6	31.2	30.7	31.1	31.3	30.6	29.0	28.5	(32)	
10%		WB	24.6	25.0	25.5	26.7	27.1	27.2	27.1	27.2	27.2	26.7	25.8	25.2	(33)	
		MCDB	27.7	28.6	29.4	30.9	31.1	30.7	30.3	30.7	30.8	29.9	28.2	27.7	(34)	
Mean Daily Temperature Range			MDBR	8.1	9.4	10.5	10.9	10.1	9.5	9.2	9.7	9.9	8.2	7.4	7.9	(35)
		5% DB	MCDBR	10.6	11.7	13.1	12.8	11.4	10.5	9.9	10.5	10.9	10.2	9.6	10.0	(36)
	MCWBR		4.9	5.0	4.9	4.6	3.9	4.0	3.7	4.0	4.2	3.9	4.1	4.7	(37)	
	5% WB	MCDBR	9.5	10.5	11.3	11.5	10.5	9.5	9.1	9.7	9.8	9.0	8.2	8.7	(38)	
		MCWBR	4.9	4.9	4.8	4.6	4.0	4.3	4.3	4.4	4.6	4.1	4.2	4.7	(39)	
Clear-Sky Solar Irradiance	taub	0.385	0.401	0.434	0.527	0.577	0.486	0.488	0.473	0.455	0.427	0.406	0.413	(40)		
	taud	2.506	2.458	2.352	2.065	1.945	2.245	2.222	2.276	2.324	2.425	2.469	2.431	(41)		
	Ebn at Noon	908	907	883	799	748	811	811	832	851	871	879	868	(42)		
	Edh at Noon	103	113	129	172	190	140	143	137	130	115	107	109	(43)		
All-Sky Solar Radiation	RadAvg	3.71	4.74	5.45	5.70	5.34	5.32	5.46	5.54	5.29	4.18	3.58	3.55	(44)		
	RadStd	0.47	0.43	0.42	0.25	0.33	0.34	0.22	0.25	0.22	0.49	0.34	0.37	(45)		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page